

Linear algebra: Exercise chapter 10

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1. Find an orthogonal matrix Q that diagonalizes $S = \begin{bmatrix} 1 & 0 & 2 \\ 0 & -1 & -2 \\ 2 & -2 & 0 \end{bmatrix}$. What is the spectral decomposition of S ?
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2. Find all orthogonal matrices that diagonalize $S = \begin{bmatrix} 9 & 12 \\ 12 & 16 \end{bmatrix}$.
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3. Consider $A = \begin{bmatrix} 2 & b \\ 1 & 0 \end{bmatrix}$. What b makes $A = Q\Lambda Q^T$ possible? What values of b will make it impossible to diagonalize A ? What values makes A singular?
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4. Find a symmetric matrix $S \in \mathbb{S}^{2 \times 2}$ such that 2 is the only eigenvalue of S . Is this matrix unique?
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5. Which of the following matrices have two positive eigenvalues? Use tests, don't compute the eigenvalues. What does it mean for a matrix to be positive definite?

$$A = \begin{bmatrix} 5 & 6 \\ 6 & 7 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & -2 \\ -2 & -5 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 10 \\ 10 & 100 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 10 \\ 10 & 101 \end{bmatrix}.$$

6. Choose a matrix M from previous exercise such that M is not positive (semi)definite. Find a vector x such that $x^T M x < 0$.
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7. Which one of the following matrix is positive (semidefinite) definite?

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & 4 \\ 4 & 3 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -5 & 2 \\ -1 & 2 & 5 \end{bmatrix}, \quad D = \begin{bmatrix} 2 & 0 & -1 \\ 0 & 3 & 2 \\ -1 & 2 & 4 \end{bmatrix}.$$

8. True or false:

- (a) Every positive definite matrix is invertible.
 - (b) A diagonal matrix with positive diagonal entries is positive definite.
 - (c) A symmetric matrix with a positive determinant is positive definite.
 - (d) Sum of two positive semidefinite matrices is positive definite.
 - (e) Difference of two positive definite matrices isn't positive definite.
 - (f) $A^T A$ is positive semidefinite or positive definite.
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9. Without multiplying $S = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix}$, find:

- (a) $\det(S)$.
 - (b) The eigenvalues of S .
 - (c) The eigenvectors of S .
 - (d) A reason why S is positive definite.
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